

Amortized: pay the simulation + training cost once, then every new observation is one cheap forward pass.

1 Simulate (offline)

prior $\theta = (\nu, A_{IC}, \varphi_{IC})$

burgers_solver

sparse noisy $\mathbf{x}_{\text{obs}}$

$\sim 10^4$ draws $(\theta, \mathbf{x})$

2 Train the flow

pairs $(\theta, \mathbf{x})$

FMPE (flow matching, zuko)

amortized $q_\phi(\theta | \mathbf{x})$

packaged as the fmpe-posterior Tesseract

3 Infer (one pass)

new observation $\mathbf{x}$

posterior samples $\theta \sim q_\phi$

marginals + SBC / TARP

single forward pass, no solver in the loop